

.NET USER GROUP KARLSRUHE · 24 SEPTEMBER 2026

# Playwright

## Beyond the Happy Path

E2E tests, accessibility and MCP in practice. Sven Hennessen · vensas GmbH

# About me

---



**Sven Hennesen**

vensas GmbH

- **Almost 20 years** of software development and architecture
- **Co-founder and co-managing director** of vensas GmbH, founded in 2023
- Consulting and development of **business software** for medium-sized companies
- Away from the computer — **hiking, cycling** or the **racing simulator**

# What changed since 2025

---

The 2025 talk showed a test runner. Playwright 1.63 is more than a test runner. These are the parts that are new for you.

1.61 – 1.63

## Control

Test locks, isolated retries and the Clock API make a test run stable and fast.

## Accessibility

ARIA snapshots and axe-core make WCAG a gate in the pipeline.

## Agents

The MCP server and the Test Agents write and repair tests.

## Demo

`npx playwright --version` — this demo uses 1.63.0. The .NET bindings are at 1.62.0.

# The demo app: Deploy or Die

A deployment game. Each deployment has a 50% success rate. After a deployment you must wait 30 seconds.

|          |                                                                                      |
|----------|--------------------------------------------------------------------------------------|
| Frontend | React 19, PrimeReact, Rspack — port 3000                                             |
| Backend  | ASP.NET Core minimal API, .NET 10 — POST /api/deploy                                 |
| Database | PostgreSQL 16, started by Docker Compose                                             |
| Tests    | TypeScript in src/deploy-or-die-frontend/tests, C# in src/deploy-or-die-dotnet-tests |

## Demo

pnpm run dev — deploy once, then watch the cooldown block the next deployment.

# Locators and auto-wait

tests/locators.spec.ts

```
// Use this order: role, label, text. These stay correct when the  
// CSS classes or the DOM structure change.  
await expect(page.getByRole('heading', { name: /Deploy or Die/i })).toBeVisible();  
await expect(page.getByLabel(/Who dares to deploy/i)).toBeEditable();  
  
// The API answer comes late, but the test does not wait manually.  
await page.getByRole('button', { name: /Deploy Now/i }).click();  
await expect(page.getByText(/Deployment successful/i)).toBeVisible();  
  
// New in 1.63: keep only the elements that the user sees.  
const visibleButtons = page.getByRole('button').visible();
```

A locator says WHAT you want. Playwright finds the element again for each action.

## Demo

pnpm run test:mocked locators — delete an await to see the failure message.

# UI mode

---

```
pnpm run test:ui
```

## What you get

- Watch mode — the test runs again after each file change
- Time travel — go back to each step and see the DOM
- Pick locator — click an element to get the locator
- Network, console and source in one window

## Demo

pnpm run test:ui — pick a locator with the mouse, then change the test and let it run again.

# Keep the tests in their own folder

Some teams put the tests next to the components, and then they set `testDir` to the source folder. This looks logical. But Playwright reads all files in `testDir` when it starts, and this needs more memory.

`playwright.config.ts`

```
// We did this. Playwright then read all of src/ at each start.  
testDir: './src',  
  
// Keep the default. The tests have their own folder.  
testDir: './tests',
```

We made this change in one project. Then we made it again in the other direction.

## What it cost us

**CI** The test job became slow. Then it stopped with a timeout.

**UI mode** The window needed some minutes to start, but not each time.

## Walkthrough

Keep `testDir` at `./tests`. A `testDir` in the source folder makes the CI job slow and the start of UI mode slow.

# Codegen

```
npx playwright codegen http://localhost:3000
```

Codegen records your clicks and writes the test. Since 1.55 it also writes assertions.

## generated

```
// Codegen gives you a start, not a finished test.
await page.getByRole('textbox', { name: 'Who dares to deploy?' }).click();
await page.getByRole('textbox', { name: 'Who dares to deploy?' }).fill('Sven');
await page.getByRole('button', { name: 'Deploy Now 🚀' }).click();
await expect(page.getByText('Deployment successful')).toBeVisible();
```

## Demo and warning

`npx playwright codegen http://localhost:3000` — record one deployment. Then correct the hard coded name and add the negative case.

# Traces

playwright.config.ts

```
use: {  
  trace: {  
    mode: 'on',  
    snapshots: { dom: true, aria: true, screen: true },  
  },  
  screenshot: 'only-on-failure',  
  video: 'retain-on-failure',  
}
```

New in 1.63: the trace records the accessibility tree and the screen, not only the DOM.

## A trace contains

- A DOM snapshot for each step, plus the ARIA tree and the screen
- All network requests and answers
- The console output and the source code

## Demo

`npx playwright show-trace test-results/.../trace.zip` — go back in time and change to the ARIA tab.

# Make the trace available in CI

A test can fail in CI and pass on your machine. Without the artifacts you can only guess. Thus the pipeline must keep them.

`.github/workflows/e2e.yml`

```
- name: Upload the report
  uses: actions/upload-artifact@v7
  if: always()
  with:
    name: playwright-report
    path: playwright-report/
    retention-days: 7

- name: Upload traces and videos
  uses: actions/upload-artifact@v7
  if: failure()
  with:
    name: playwright-traces
    path: test-results/
    retention-days: 7
```

The report goes up after each run. The trace and the video go up only after a failure.

## Walkthrough

Add the two steps to your workflow. The first one runs always. The second one runs only after a failure.

# Keep the test pyramid

Codegen, UI mode and the agents make an E2E test look cheap. It is not. An E2E test needs more time to write and to keep than a unit test.

## From a team that I mentored

The team wrote mostly Playwright tests. React Testing Library was sufficient for 80% of these scenarios. The suite became very large, and then the CI runners became unreliable.

## What I take from this

Many

Unit and component tests. They are fast and cheap.

Few

E2E tests. Use them for the important paths.

An unreliable E2E test is worse than no E2E test.

## Walkthrough

Count the E2E tests in your project. Then ask which of them a unit test or a component test can do.

# Mock the API

tests/deploy-failure.spec.ts

```
// Set the route before you go to the page.
await page.route('**/api/deploy', async (route) => {
  await route.fulfill({
    status: 200,
    contentType: 'application/json',
    body: JSON.stringify({
      success: false,
      message: '💣 BOOM! The database just went on vacation.',
      deployedBy: 'Unlucky Developer',
    }),
  });
});

await page.goto('/');
```

You control the answer. Thus you can test the error path, which occurs only 50% of the time.

## Demo

pnpm run test:mocked — 17 tests in about 10 seconds, with no backend and no database.

# The real backend

tests/testcontainers.ts

```
const composeEnvironment = await new DockerComposeEnvironment(  
  composeFilePath,  
  'docker-compose.integration.yml',  
) .up();  
  
const backend = composeEnvironment.getContainer('backend-1');  
process.env.API_URL = `http://localhost:${backend.getMappedPort(5000)}`;
```

Docker Compose is more stable than a container that you build in the test code.

## Mock or real

**Mock**

Fast and stable. Blind to the backend.

**Real**

Finds integration faults. Slow.

Use mocks for the many tests. Use the real backend for the few important paths.

## Demo

pnpm run test:integration — the containers start, then 4 tests run against the real API.

# Start the dependencies with Compose

## What broke when I started each container in the test code

- The start order
- The readiness check before the next container starts
- The network between the containers and the port mapping
- Containers that stayed after the run

tests/docker-compose.integration.yml

```
postgres:
  healthcheck:
    test: ["CMD-SHELL", "pg_isready -U postgres"]

backend:
  depends_on:
    postgres:
      condition: service_healthy
```

depends\_on with a healthcheck replaces the wait code in the test.

## Walkthrough

Put the healthcheck and depends\_on in the Compose file. Then the test code needs no wait code.

# Test locks — new in 1.63

All integration tests write to the same database. Before 1.63 you made the whole file serial, or you accepted the flakiness.

tests/deploy-integration.spec.ts

```
test.describe('Integration Tests', { lock: 'deploy-db' }, () => {
  test('should deploy against the real backend', async ({ page }) => {
    // ...
  });

  test('the API rejects a request without a name', async ({ request }) => {
    // ...
  });
});

// The other tests keep their parallel speed.
```

Two tests with the same lock name never run at the same time — across files and across workers. Playwright Test only — the .NET bindings have no lock, because xUnit is the runner there.

## Demo

Run the integration project with `--workers=4`. The Timeline in the report shows that the four locked tests do not overlap.

# Speedboard and Timeline

The mocked tests need 10 seconds. The integration tests need more than 4 minutes. The report shows you where the time goes.

```
pnpm run test:report
```

## Two tabs in the HTML report

**Speedboard** A list of all tests, sorted by duration. The slowest test is at the top.

**Timeline** The wall clock time of the run. It shows where a worker waits.

Most of our 4 minutes is the container build, not the tests.

## Demo

pnpm run test:report after the integration run — look at Speedboard, then at Timeline.

# Isolated retries — new in 1.62

playwright.config.ts

```
export default defineConfig({
  retries: process.env.CI ? 2 : 0,

  // New in 1.62.
  retryStrategy: 'isolated',
});
```

The retries run one after the other at the end of the run, not immediately under full load. Playwright Test only — not available in the .NET bindings.

## Why this helps

A test that fails only because of parallel load now passes on the isolated retry. This tells you the cause: it is the load, not the code.

## Demo

`pnpm run test:retries` — the red test retries. With isolated both retries run last, back to back. Comment the line out and run again: the first retry then runs in the middle of the suite.

# Mock or real — how I choose

The two approaches are not in competition. Both are correct, and both add value. I choose with the complexity of the project.

## Three questions before I decide

**Services** How many services does the scenario touch?

**Contract** How complex is the API contract?

**Test** How complex is the test itself?

I answer these questions again for each project. Your environment and your test plan give the answer, not a rule.

## Walkthrough

Answer the three questions for your own project. The demo app has one service and a small contract.

# The E2E tests are not only a frontend job

Playwright tests a user interface. Thus teams put the tests in the frontend pipeline only. But a test against a real backend image also tests the backend.

## What happened to a team that I mentored

The team released an API that was not compatible. Nobody ran the E2E tests before the release.

## What I recommend

- Run the tests after each frontend build
- Run them after each backend build too
- Or run them nightly — your test plan decides

## Walkthrough

Find out where your E2E tests run today. Then add the backend build as a trigger, or run them nightly.

# The Clock API

The app blocks a second deployment for 30 seconds. A test must not wait 30 seconds.

tests/cooldown.spec.ts

```
await page.clock.install();
await page.goto('/');

await page.getByRole('button', { name: /Deploy Now/i }).click();
await expect(page.getByRole('button', { name: /Next deploy in 30s/i })).toBeVisible();

// Move the time forward by 5 seconds and let the timers run.
await page.clock.runFor(5_000);
await expect(page.getByRole('button', { name: /Next deploy in 25s/i })).toBeVisible();

// Skip the full cooldown. The test needs milliseconds, not 30 seconds.
await page.clock.fastForward('00:30');
await expect(page.getByRole('button', { name: /Deploy Now/i })).toBeEnabled();
```

install() must come before every other clock call.

## Demo

pnpm run test:clock — 3 tests in about 2 seconds, for a cooldown of 30 seconds.

# Accessibility as a gate

tests/accessibility.spec.ts

```
const results = await new AxeBuilder({ page })
  .withTags(['wcag2a', 'wcag2aa', 'wcag21a', 'wcag21aa', 'wcag22aa'])
  .analyze();

await testInfo.attach('axe-results.json', {
  body: JSON.stringify(results.violations, null, 2),
  contentType: 'application/json',
});

expect(results.violations.map((v) => ({ rule: v.id, impact: v.impact }))).toEqual([]);
```

The European Accessibility Act applies since June 2025. Thus this check belongs in the pipeline.

## Be honest about the limit

An automatic scan finds about 30% to 50% of the WCAG problems. It does not replace a test with a screen reader.

## Demo

pnpm run test:a11y fails with button-name, color-contrast and label. Set ACCESSIBLE\_MODE = true in src/a11yMode.ts and it passes.

# ARIA snapshots

```
tests/accessibility.spec.ts
```

```
await expect(page.locator('.card-header')).toMatchAriaSnapshot(`
  - heading /Deploy or Die/ [level=1]
  - paragraph: /Every deployment is a gamble/
`);
```

The snapshot asserts the accessibility tree, not the DOM. It shows roles and names, not CSS.

## Why this is different from axe-core

**axe-core** Finds known WCAG defects, for example too little contrast

**ARIA snapshot** Finds a change of the structure, for example a lost role or a wrong heading level

The two checks find different faults. Use both.

## Demo

Change the h1 to an h2. The axe test still passes, but the ARIA snapshot fails.

# The same test in C#

DeployTests.cs

```
public class DeployTests : PageTest
{
    [Fact]
    public async Task Cooldown_BlocksTheSecondDeployment()
    {
        await Page.Clock.InstallAsync();
        await Page.GotoAsync("http://localhost:3000");

        var deploy = Page.GetByRole(AriaRole.Button,
            new() { NameRegex = new Regex("Deploy Now") });
        await deploy.ClickAsync();

        // Move the time forward instead of a wait of 30 seconds.
        await Page.Clock.FastForwardAsync("00:30");
        await Expect(deploy).ToBeEnabledAsync();
    }
}
```

The API is equal in all languages. Only the names use the .NET style. The package is Microsoft.Playwright.Xunit.v3 1.62.0 — the bindings follow Node, one version behind. NUnit and MSTest also exist.

## Demo

dotnet test — 4 tests pass. Compare the C# file with the TypeScript file.

# What the .NET bindings give you

The browser API is the same. The test runner is not — in .NET your runner is xUnit, NUnit or MSTest, not Playwright Test.

## Works in C#

|                       |                                        |
|-----------------------|----------------------------------------|
| <b>Locators</b>       | GetByRole, GetByLabel, auto-wait       |
| <b>Clock API</b>      | InstallAsync, FastForwardAsync         |
| <b>ARIA snapshots</b> | ToMatchAriaSnapshotAsync               |
| <b>Mocking</b>        | RouteAsync and the API request context |
| <b>Traces</b>         | Recorded, then read with show-trace    |
| <b>Codegen</b>        | playwright codegen --target csharp     |

## Playwright Test only

|                         |                                 |
|-------------------------|---------------------------------|
| <b>Test locks</b>       | Use an xUnit collection instead |
| <b>Isolated retries</b> | The runner controls the retries |
| <b>UI mode</b>          | No equivalent                   |
| <b>Speedboard</b>       | Use your own test report        |
| <b>Test Agents</b>      | They write TypeScript           |

## Demo

`npx playwright codegen --target csharp-xunit http://localhost:3000` — the recorder writes C#, not TypeScript.

# Accessibility in C#

AccessibilityTests.cs

```
var results = await Page.RunAxe(new AxeRunOptions
{
    RunOnly = new RunOnlyOptions
    {
        Type = "tag",
        Values = ["wcag2a", "wcag2aa", "wcag21a", "wcag21aa", "wcag22aa"],
    },
});

Assert.Empty(results.Violations.Select(v => v.Id).ToArray());
```

Deque.AxeCore.Playwright 4.13.0 adds RunAxe() to Page and Locator. It uses the same axe-core engine as the TypeScript package.

## Same engine, same result

Both languages report the same three rules on our form: button-name, color-contrast and label. Accessibility is not a JavaScript topic.

## Demo

dotnet test — 5 tests, 1 red. The C# result names the same three rules as the TypeScript result.

# The Playwright MCP server

The MCP server gives an AI agent a real browser. Since 1.62 the server is part of Playwright.

```
npx playwright mcp
```

No separate installation is necessary.

## Why it is different from a screenshot agent

|               |                                                             |
|---------------|-------------------------------------------------------------|
| <b>Input</b>  | The agent gets the accessibility tree, not pixels           |
| <b>Result</b> | Fewer tokens, and the same answer each time                 |
| <b>Effect</b> | The agent writes role based locators, because it sees roles |

## Demo

Give the agent this task: explore <http://localhost:3000>. It finds the cooldown without a hint.

# Test Agents

`NPX PLAYWRIGHT INIT-AGENTS --LOOP=CLAUDE`

Make the agent definitions again after each Playwright update, to get the new tools.

## STEP 1

### Planner

Explores the app and writes a test plan in Markdown to specs/.



## STEP 2

### Generator

Makes test files from the plan and proves each locator against the running app.



## STEP 3

### Healer

Runs a failed step again, looks for the element that replaced the old one and proposes a patch.

## Demo

Change the button text from 'Deploy Now' to 'Ship It'. Let the healer repair the test.

# Where you must hold the reins

## A generated test can look correct and be worthless

Typical defects: a hard coded wait, a CSS selector instead of a role, and an assertion that is always true.

```
// Generated – and wrong.  
await page.waitForTimeout(3000);  
await page.click('.p-button.p-component.deploy-button');  
await expect(page.locator('body')).toBeVisible();
```

This test passes. It tests nothing.

## The rule that works for us

Agent

Explores and writes the first draft.

Human

Reviews, then puts it in the pipeline.

## Demo

Find the three defects in the listing above. Then compare them with `tests/locators.spec.ts`, the reviewed version of the same test.

# Lessons learned

---

1

## **Divide the tests into projects** MOCKED · INTEGRATION

The mocked project needs no container. Thus a developer gets an answer in 10 seconds.

2

## **Use the real backend only for the important paths**

Each integration test costs minutes. Four of them are enough for us.

3

## **Start the containers with Compose, not in the test code**

A global setup with Docker Compose is much more stable than a container per test.

4

## **Switch on traces in the pipeline**

A trace answers the question 'why did it fail in CI' without a second run.

5

## **Make accessibility a gate on day one**

Three defects in a small form. A large application has hundreds.

# Not in this talk

---

1.61

## WebAuthn passkeys

A virtual authenticator tests a login with a passkey, with no hardware.

1.62

## Component testing

A new model with stories and galleries mounts one component in a real browser.

1.48

## WebSocket mocking

`routeWebSocket()` replaces a realtime backend, for example a live deployment log.

### Also worth a look

- `storageState` with IndexedDB — keep a login between tests
- `npx playwright trace` — read a trace in the terminal, with no window
- Copy prompt in the report — give a failed test to an LLM with one click

# Questions

---

THANK YOU

## Questions?

The demo project and these slides are in the repository. Write to me if you want the link.

[kontakt@vensas.de](mailto:kontakt@vensas.de)

[vensas.de](https://vensas.de)